Software Engineering in Safety and Mission Critical Applications – Research Trends and Highlights of Information Technology in Automation group at Aalto University

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Background and Related Works

Safety and mission-critical software systems are essential in domains such as robotics and industrial automation, energy, transportation, aerospace, and the defence industry, where software failures may lead to severe safety, operational, or economic consequences. Engineering such systems requires rigorous requirements specification, hazard analysis, verification and validation, and traceability, which is also one reason why these sectors are more lagging behind in the realm of AI in software engineering compared to traditional software sector. Recent advances in the Artificial Intelligence (AI), particularly Large Language Models (LLMs), offer new opportunities to support and automate software engineering activities, including requirements analysis, software design, code generation, testing, and safety assurance.

Recent research has demonstrated the potential of Large Language Models (LLMs) to support various software engineering activities. Ferrari and Spoletini [1] explored the integration of LLMs and formal methods in requirements engineering, highlighting opportunities to improve the trustworthiness of AI-generated specifications. Ma et al. [2] further demonstrated LLM-assisted translation of natural-language requirements into Linear Temporal Logic (LTL), achieving promising results while revealing remaining limitations in semantic accuracy. In safety engineering, Qi et al. [3] investigated LLM-assisted Systems-Theoretic Process Analysis (STPA), demonstrating its potential for hazard identification while emphasising the need for human expertise. Dokas [4] identified limitations in existing LLM evaluation benchmarks, particularly regarding causal reasoning, safety-specific knowledge, and consistency. Furthermore, Sivakumar et al. [5] and Odu et al. [6] investigated LLM-assisted safety-case generation, demonstrating the feasibility of automating structured assurance arguments while highlighting limitations in their accuracy and interpretation. These studies demonstrate progress in AI-assisted engineering and safety assurance. However, introducing AI into the development lifecycle still presents significant challenges and research gaps.

Current Research Challenges and Open Problems

Despite recent advances in AI-assisted software engineering, several fundamental challenges remain especially in safety and mission-critical software systems:

1. Correctness and Trustworthiness: AI-generated requirements, code, and test cases may contain inaccuracies, inconsistencies, or omissions. Ensuring semantic correctness, compliance with safety requirements remains an open challenge.

2. Formal Methods in AI assisted engineering: Existing formal methods provide rigorous verification capabilities, but integrating them effectively with AI-assisted development workflows remains challenging.

3. Traceability and Continuous Safety Assurance: Maintaining consistency and traceability between safety requirements, system architectures, implementations, verification results, and assurance arguments remains labour-intensive.

References

[1] A. Ferrari and P. Spoletini, “Formal requirements engineering and large language models: A two-way roadmap,” Information and Software Technology, vol. 181, Art. no. 107697, 2025

[2] Z. Ma et al., “Bridging natural language and formal specification—Automated translation of software requirements to LTL via hierarchical semantics decomposition using LLMs,” in Proc. 40th IEEE/ACM International Conference on Automated Software Engineering, 2025

[3] Y. Qi, X. Zhao, S. Khastgir, and X. Huang, “Safety analysis in the era of large language models: A case study of STPA using ChatGPT,” Machine Learning with Applications, Art. no. 100622, 2025

[4] I. M. Dokas, “From hallucinations to hazards: Benchmarking LLMs for hazard analysis in safety-critical systems,” Safety Science, vol. 194, Art. no. 107056, 2026.

[5] M. Sivakumar, A. B. Belle, J. Shan, and K. Khakzad Shahandashti, “Prompting GPT-4 to support automatic safety case generation,” Expert Systems with Applications, vol. 255, Art. no. 124653, 2024

[6] O. Odu, A. B. Belle, S. Wang, S. Kpodjedo, T. C. Lethbridge, and H. Hemmati, “Automatic instantiation of assurance cases from patterns using large language models,” Journal of Systems and Software, vol. 222, Art. no. 112353, 2025

[7] M. Xavier, T. Laikh, S. Patil and V. Vyatkin, "LLM-Powered Multi-Actor System for Intelligent Analysis and Visualization of IEC 61499 Control Systems," IECON 2024 - 50th Annual Conference of the IEEE Industrial Electronics Society, Chicago, IL, USA, 2024

[8] P. Ovsiannikova, T. Liakh, P. Jhunjhunwala and V. Vyatkin, "Generative AI Co-Pilot for Rapid Prototyping of IEC 61499 Control Applications," 2024 IEEE 29th International Conference on Emerging Technologies and Factory Automation (ETFA), Padova, Italy, 2024,

[9] Ovsiannikova, Polina, et al. "Towards a Trustworthy Co-pilot for IEC 61499 Function Block Composition." International Workshop on Service Oriented, Holonic and Multi-Agent Manufacturing Systems for Industry of the Future. Cham: Springer Nature Switzerland, 2025.

[10] A. King and V. Vyatkin, "LLM-based Iterative Refinement of Finite-State Machines with STPA Controller Constraints and Generation of IEC 61499 Code," 2025 IEEE 30th International Conference on Emerging Technologies and Factory Automation (ETFA), Porto, Portugal, 2025

[11] T. Lyu, U. D. Atmojo and V. Vyatkin, "LLM4VC: Harnessing Large Language Models for Virtual Commissioning of IEC 61499 Automation Systems," 2025 IEEE 34th International Symposium on Industrial Electronics (ISIE), Toronto, ON, Canada, 2025.

[12] E. Shahinas, A. King and U. D. Atmojo, "Safe reinforcement learning for ship energy management optimization with LLM-based reward shaping," 2025 IEEE 30th International Conference on Emerging Technologies and Factory Automation (ETFA), Porto, Portugal, 2025.

[13] A. King, E. Shahinas, U. D. Atmojo and V. Vyatkin, "Exploring Safe Reinforcement Learning Using Safety Shields Derived With System-Theoretic Process Analysis: A Case-Study on a Cruise Ship Hotel System," 2025 IEEE 30th International Conference on Emerging Technologies and Factory Automation (ETFA), Porto, Portugal, 2025

Highlights of Recent Developments from the Group

The Information Technology in Automation research group investigates enablers of future industry systems, from the industrial manufacturing, continuous processes, energy management, and human-centric perspectives. One key research area of the group is software engineering methods and measures for safety critical industrial systems – such as those used in robotics and industrial automation systems, and in the Instrumentation & Control (I&C) system used in nuclear power plant.

Some highlights of the work done in the group include:

1) Multi agent framework that supports the developers in code analysis and testing of industrial automation system [7]

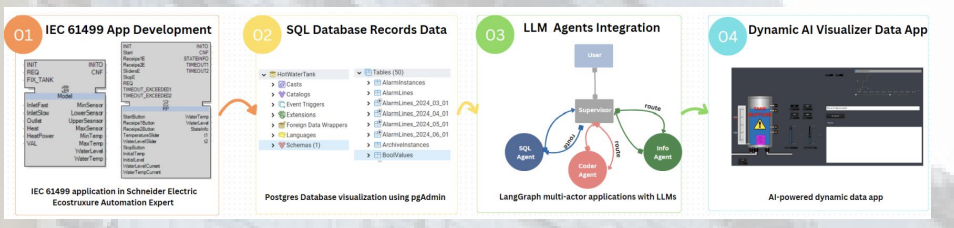


Figure 1. LLM multi actor framework for 1)

2) Co-pilot that support developers in rapid prototyping of industrial automation system written in IEC 61499 standard language [8], then enhanced with formal model checking feature [9]

3) FBAssistant - framework for LLM-enabled code generation for industrial automation system software through iterative prompting, augmented with Systems-Theoretic Process Analysis (STPA) safety analysis [10]

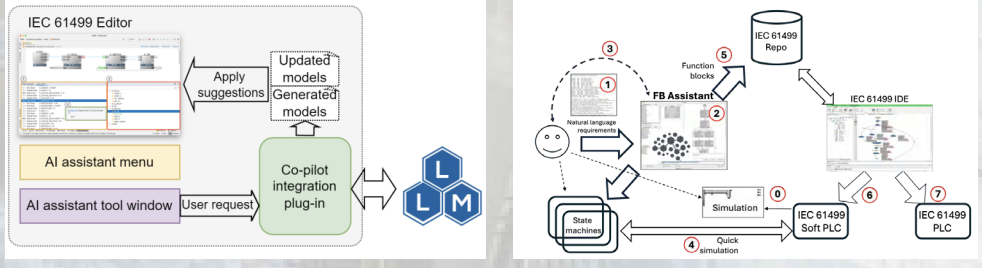


Figure 2. Copilot system for 2) (left), FBAssistant iterative prompting framework for 3) (right)

4) LLM4VC – cloud-based LLM-enabled validation agent for industrial automation software [11]

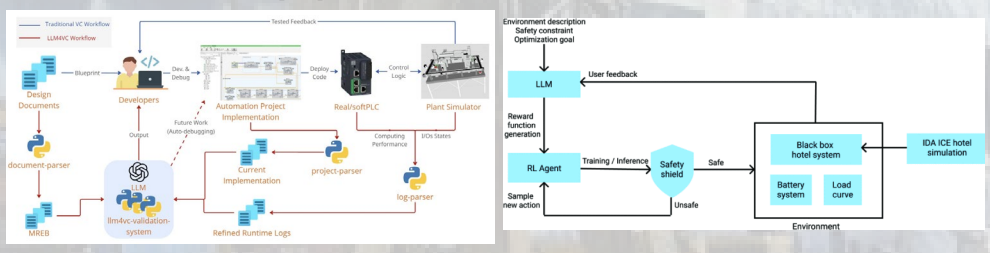
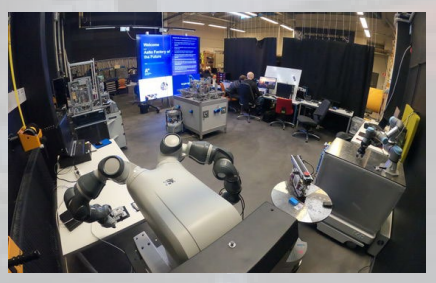


Figure 3. LLM4VC framework for 4) (left), workflow for the development of safe reinforcement learning agent for 5) (right)

5) LLM-enabled framework for the development of safe reinforcement learning based AI agent based on reward shaping, safety shielding, and Systems-Theoretic Process Analysis (STPA) safety and hazard analysis [12], [13]

The group experiments and validates most of their research work in a highly realistic industrial testbed Aalto Factory of the Future at Aalto University. See more



See more about the group

Figure 4. Aalto Factory of the Future industrial testbed